

# Page-Cache Corruption Primitives: Evolution of Linux Kernel Local Privilege Escalation (2016–2026)

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**Abstract**—This paper presents a systematic analysis of five Linux kernel local privilege escalation (LPE) vulnerabilities spanning 2016 to 2026: Dirty COW (CVE-2016-5195), Dirty Pipe (CVE-2022-0847), Copy Fail (CVE-2026-31431), Dirty Frag (CVE-2026-43284/43500), and Fragnesia (CVE-2026-46300). We identify a structural evolutionary pattern: the attack surface has migrated from the Virtual Memory subsystem through File System pipes to the Crypto API and Networking (XFRM/IPsec) stacks. Exploit reliability has improved from probabilistic race-condition exploitation to fully deterministic, cross-distribution primitives requiring no kernel offsets. We deploy a Docker-based research testbed with eBPF instrumentation to characterize each vulnerability’s page-cache corruption primitive, evaluate mitigation effectiveness, and propose detection signatures. We further analyze the role of AI-assisted vulnerability discovery in reducing time-to-disclosure from weeks to approximately one hour. Our findings have significant implications for Linux kernel security hardening, container security, and the emerging AI-augmented threat landscape.

**Index Terms**—Linux kernel security, local privilege escalation, page-cache corruption, CVE analysis, container security, eBPF, AI-assisted vulnerability discovery

## I. INTRODUCTION

The Linux kernel underpins the majority of cloud infrastructure, containerized workloads, and production server deployments globally. Local privilege escalation vulnerabilities represent a persistent and critical threat class, enabling attackers who have obtained initial access — through phishing, RCE, or supply chain compromise — to escalate to root and achieve complete system compromise. Recent months have witnessed an unprecedented cluster of high-impact Linux kernel LPE disclosures, including Copy Fail (April 2026), Dirty Frag (May 2026), and Fragnesia (May 2026), all sharing a common page-cache corruption mechanism but arising in distinct kernel subsystems.

This work asks: what structural pattern connects these vulnerabilities, and what does their evolution reveal about gaps in Linux’s defensive architecture? We make the following contributions: (1) the first unified analysis of five page-cache LPEs across a decade, (2) a reproducible Docker testbed for kernel vulnerability research, (3) a taxonomy of mitigations and their effectiveness, and (4) an analysis of AI’s emerging role in vulnerability discovery.

## II. BACKGROUND: LINUX PAGE CACHE ARCHITECTURE

The Linux page cache serves as the in-memory representation of file-backed data, enabling efficient I/O by caching disk pages in RAM. Kernel subsystems — including the Virtual Memory manager, pipe buffers, crypto sockets (AF\_ALG), and networking stacks — interact with the page cache through controlled interfaces governed by Copy-on-Write (CoW) semantics. A recurring vulnerability pattern in the 2016–2026 period exploits logic flaws that allow unauthorized writes into page-cache-backed pages of read-only files, bypassing CoW protections.

## III. TECHNICAL ANALYSIS OF EACH VULNERABILITY

### A. Dirty COW (CVE-2016-5195, 2016)

*Affected subsystem:* Virtual Memory — copy-on-write race condition in `mm/memory.c`

Dirty COW exploits a race condition in the kernel’s copy-on-write VM path via `/proc/self/mem`, allowing an unprivileged user to write to read-only memory mappings. The exploit requires winning a race window and is notoriously slow and unstable, sometimes crashing the system. Despite its unreliability (~30% success rate per attempt), Dirty COW remained undetected for 9 years before disclosure.

### B. Dirty Pipe (CVE-2022-0847, 2022)

*Affected subsystem:* File System — uninitialized `PIPE_BUF_FLAG_CAN_MERGE` in `fs/pipe.c`

Dirty Pipe achieves 100% reliable LPE by exploiting uninitialized pipe buffer flags. A `splice()` call can introduce a page-backed entry into a pipe with the `CAN_MERGE` flag erroneously set, enabling overwrite of arbitrary read-only page-cache content. Version-specific to Linux 5.8+, Dirty Pipe marked the transition to deterministic exploitation.

### C. Copy Fail (CVE-2026-31431, 2026)

*Affected subsystem:* Crypto API — logic flaw in `algif_aead` module (AF\_ALG interface)

Copy Fail is a straight-line logic flaw in the kernel’s userspace crypto API. An unprivileged process opens an AF\_ALG AEAD socket and uses `splice()` to induce a deterministic 4-byte write into the page cache of any readable file. A single 732-byte Python script achieves root on all major

distributions since 2017. Notably, it was discovered by AI system Xint Code in approximately one hour.

#### D. Dirty Frag (CVE-2026-43284 / CVE-2026-43500, 2026)

*Affected subsystem:* Networking — IPsec ESP + RxRPC CoW bypass in xfrm\_input.c

Dirty Frag chains two CVEs: a flaw in ESP fragment reassembly (CVE-2026-43284) and an RxRPC interaction (CVE-2026-43500). Together they allow page-fragment CoW bypass via the networking stack, enabling arbitrary page-cache writes from unprivileged context. This marked the first time networking subsystems became the LPE attack surface.

#### E. Fragnesia (CVE-2026-46300, 2026)

*Affected subsystem:* Networking — XFRM ESP-in-TCP skb coalescing flaw (skb\_try\_coalesce)

Fragnesia was ironically activated by the Dirty Frag patch itself. A missing SKBFL\_SHARED\_FRAG flag propagation in skb\_try\_coalesce() allows the kernel to treat shared, file-backed pages as privately writable. The public PoC overwrites /usr/bin/su to gain root. Discovered with Zellic’s AI auditing tool, Fragnesia exemplifies how security patches can inadvertently expand the attack surface.

### IV. COMPARATIVE EVALUATION

#### V. KEY FINDINGS

**F1 — Reliability Evolution:** Exploit reliability improved from probabilistic (~30%, Dirty COW) to fully deterministic (100%, all 2026 CVEs). This eliminates a key operational barrier for attackers.

**F2 — Subsystem Migration:** The attack surface migrated: VM (2016) → FS (2022) → Crypto API / Networking (2026). Each migration exploited subsystems with less security scrutiny than the prior target.

**F3 — Unified Primitive:** All five CVEs share page-cache corruption as the underlying primitive, suggesting that a unified page-cache write auditing layer (e.g., via eBPF LSM hooks) could detect all variants.

**F4 — AI Discovery Acceleration:** Copy Fail was found in ~1 hour by Xint Code AI; Fragnesia by Zellic AI. Manual discovery timelines measured weeks. AI is fundamentally accelerating the vulnerability lifecycle.

**F5 — Patch-Induced Exposure:** Fragnesia was accidentally introduced by the Dirty Frag patch. Security patches themselves must be treated as a source of new attack surface requiring adversarial review.

### VI. MITIGATION TAXONOMY

#### VII. FUTURE WORK

We identify three primary directions for future research. First, development of a unified page-cache integrity monitor using eBPF LSM hooks that can detect all variants of the page-cache corruption primitive at runtime, independent of the triggering subsystem. Second, formal verification of critical kernel subsystems — particularly AF\_ALG, XFRM, and pipe buffer management — using model checkers such as CBMC or

Kani. Third, systematic study of AI-assisted kernel auditing to understand which bug classes are most amenable to automated discovery and what guardrails are needed for responsible AI-driven vulnerability research.

### VIII. CONCLUSION

The 2016–2026 period reveals a clear evolutionary arc in Linux kernel LPE exploitation: from slow, unreliable race conditions to deterministic, cross-distribution primitives exploiting networking and crypto subsystems. The common thread — page-cache corruption — provides both an explanatory framework and a defensive target. As AI tools accelerate vulnerability discovery, the kernel security community must invest in proactive AI-augmented auditing, formal verification, and runtime integrity enforcement. This research provides the first unified treatment of this exploit family and a reproducible testbed for future investigation.

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TABLE I  
COMPARATIVE ANALYSIS OF LINUX KERNEL LPE CVEs (2016–2026)

| CVE            | Name       | Year | Subsystem     | Race-Free | Container Escape | AI Found | CVSS |
|----------------|------------|------|---------------|-----------|------------------|----------|------|
| CVE-2016-5195  | Dirty COW  | 2016 | VM / mm       | No        | No               | No       | 7.8  |
| CVE-2022-0847  | Dirty Pipe | 2022 | FS / pipe     | Yes       | No               | No       | 7.8  |
| CVE-2026-31431 | Copy Fail  | 2026 | Crypto AF_ALG | Yes       | Yes              | Yes      | 7.8  |
| CVE-2026-43284 | Dirty Frag | 2026 | Net XFRM/ESP  | Yes       | Yes              | No       | 7.8  |
| CVE-2026-46300 | Fragnesia  | 2026 | XFRM ESP-TCP  | Yes       | Yes              | Yes      | 7.8  |

TABLE II  
MITIGATION EFFECTIVENESS BY CVE

| Mitigation         | Dirty COW | Dirty Pipe | Copy Fail | Dirty Frag | Fragnesia |
|--------------------|-----------|------------|-----------|------------|-----------|
| Kernel Patch       | Full      | Full       | Full      | Full       | Full      |
| Module Blacklist   | N/A       | N/A        | Full      | Full       | Full      |
| SMEP/SMAP          | Partial   | No         | No        | No         | No        |
| SELinux/AppArmor   | Partial   | Partial    | No        | No         | No        |
| Seccomp-BPF        | Partial   | Partial    | Partial   | Partial    | Partial   |
| eBPF LSM Detection | Yes       | Yes        | Yes       | Yes        | Yes       |